MEET VYAS

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EDUCATION

Ahmedabad University — Gujarat, India
B.Tech. in Mechanical Engineering with Minors in Physics and Mathematics (GPA - 3.28/4) — *August 2020 – May 2024*

PUBLICATIONS

- Brauer, Kaley and Dash, Aditya Prasad and Vyas, Meet J. and Massala, Ahmed Salim Stiven Briand. "From Simulations to Surveys: Domain Adaptation for Galaxy Observations".
  Machine Learning and the Physical Sciences Workshop @ NeurIPS 2025. 2025
- Terry, Sean K. et al. (*incl.* Vyas, M. J.). "Predictions of the Nancy Grace Roman Space Telescope Galactic Exoplanet Survey. IV. Lens Mass and Distance Measurements".
  arXiv:2510.13974. 2025
- Di Valentino, Eleonora et al. (*incl.* Vyas, M. J.). "The CosmoVerse White Paper: Addressing Observational Tensions in Cosmology with Systematics and Fundamental Physics".
  Phys. Dark Univ. 48, 101965. 2025
- Bambhaniya, Parth and Vyas, Meet J. and Joshi, Pankaj S. and de Gouveia Dal Pino, Elisabete M. "Retrograde Precession of Relativistic Orbits and the Quest for Charged Black Holes".
  Phys. Dark Univ. 48, 101949. 2025
- Andreoni, Igor et al. (*incl.* Vyas, M. J.). "Rubin ToO 2024: Envisioning the Vera C. Rubin Observatory LSST Target of Opportunity Program".
  arXiv:2411.04793 (Submitted to Astrophysical Journal Supplement Series). 2024
- Vyas, M. J. et al. "Experimental Investigation of a Concentrated Photovoltaic System Integrated With Water Sprinkler Generating Electricity and Hot Water".
  ASME GTINDIA2023, DOI: 10.1115/GTINDIA2023-118667. 2023

CONFERENCE AND POSTER TALKS

National Conference ARYA-2025 (Hybrid Mode) Poster Competition — SESTECH, Gujarat University
Theme - Statistics, AI&ML, — *March 2025*

Gas Turbine Conference 2023 — American Society of Mechanical Engineers
Theme - Gas Turbine and Renewables — *December 2023*

SUMMER SCHOOLS

IAIFI Summer School 2025 — MIT
Theme - AI and Statisitics in Physics — *August 2025*

IAIFI Summer School 2024 — MIT
Theme - AI and Statisitics in Physics — *August 2024*

Zwicky Transient Facility (ZTF) Summer School 2024 — University of Minnesota
Theme - AI in Cosmology — *July 2024*

Gravitational-Wave Astronomy — International Center for Theoretical Sciences, TIFR
Theme - Gravitational Wave Astronomy — *July 2024*

Fifth IUCSS Summer School — Indiana University, Bloomington
Theme - Lorentz- and CPT-violating Standard-Model Extension — *May 2024*

Zwicky Transient Facility (ZTF) Summer School 2023 — University of Minnesota
Theme - Transients — *July 2023*

Technical Skills

Programming & Scripting: C/C++, Fortran, Python, R, MATLAB, Mathematica, Wolfram Language
Astronomy, Cosmology & Inference: Astropy, RTModel, PyLIMA, MulensModel, sncosmo, redback, Cobaya, CAMB, CLASS, DiscoDJ, CosmoMC
Statistical & Bayesian Inference: emcee, Bilby, dynesty, PyMC, Stan, corner.py, GetDist
Scientific Computing & HPC: CUDA, MPI, OpenMP, SLURM, OpenFOAM, Simulink, LabVIEW, LTspice
Machine Learning & Data Analysis: NumPy, SciPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Weights & Biases
Large-Scale Data Processing: Dask, HDF5, FITS (cfitsio), PySpark
Numerical Methods & Optimization: SciPy Optimize, JAX, Autograd, Numba
Development & Reproducibility: Git, GitHub, GitLab, Docker, Conda, Make/CMake, Jupyter Notebook, pip
Testing & Debugging: PyTest, Valgrind, Wireshark
Scientific Writing & Visualization: LaTeX, BibTeX, RevTeX, Matplotlib, Seaborn, Markdown

Experience

Research Intern — May 2024 – January 2026
International Centre for Space and Cosmology, Ahmedabad University — *Gujarat, India*

Undergraduate Teaching Assistant — October 2023 – March 2024
Ahmedabad University — *Gujarat, India*

Summer Intern — May 2023 – July 2023
Physical Research Laboratory — *Gujarat, India*

Physics Journal Club Coordinator — June 2021 – June 2022
Ahmedabad University — *Gujarat, India*

Student Ambassador — June 2021 – January 2022
Ahmedabad University — *Gujarat, India*

Undergraduate Peer Tutor — August 2021 – December 2021
Ahmedabad University — *Gujarat, India*

Awards and Honors

IAIFI Summer School 2025 Hackathon Winner: Best Group Effort	2025
We Are All Humans (WAAH) Research Intern Fellowship	2025
Outstanding Performance and Impact on Community Award by Ahmedabad University	2024
Travel Grant by the Heising Simons Foundation for the Rubin ToO workshop	2024
Travel Grant by Ahmedabad University for presenting my research at the ASME Gas Turbine India Conference 2023	2023

International Collaborations

Member of The Roman Science Collaboration — October 2025 – Present
Member of The COST Action Cosmoverse Working Groups 1,2, and 3 — August 2025 – Present
Member of The Wide-Field spectroscopic telescope Science Team — July 2025 – Present
Member of The Roman Galactic Exoplanet Survey Project Infrastructure Team (RGES PIT) — March 2025 – Present
Member of Roman Wide Field Instrument Working Group — August 2024 – Present
Member of the LSST Transient and Variable Stars (TVS) Science Collaboration — July 2024 – Present
Member of the Cosmic Explorer Consortium — July 2024 – Present
Member of Rubin Undergraduate Network — July 2024 – Present

Workshops Organized

Roman Microlensing Data Challenge 25 AAS Workshop — RGES PIT
Theme - Microlensing — *January 2026*

Fourth Intermediate Python for Astronomical Software Development — TVS Science Collaboration LSST
Theme - HPC, Parallelization, mpi4py, CUDA — *August 2025*

Fourth TVS Software Workshop — TVS Science Collaboration LSST
Theme - Science cases and practical aspects of using LSST data using Software — *March 2025*

Third Intermediate Python for Astronomical Software Development — TVS Science Collaboration LSST
Theme - Unit Testing, Github Actions, Code Profiling — *December 2024*

AI and Cosmology: Frontiers for the next decade 2024 — ICSC, Ahmedabad University
Theme - AI and ML in Cosmology — *April 2024*

WORKSHOPS ATTENDED

PhysCOS Second Early Career Workshop — NASA

Observatories Forum Workshop — AEON+ Collaboration
Theme - Time Domain and Multimessenger Astrophysics — *September 2025*

Rubin Community Workshop 2025 — Legacy Survey of Space and Time
Theme - Rubin Science — *July 2025*

Shaping the Future of TIME DOMAIN ASTRONOMY in the Rubin Era — TVS Science Collaboration LSST
Theme - Time Domain Astronomy — *May 2025*

Spectroscopy Calibration Workshop — Roman Wide Field Instrument Calibration Working Group
Theme - Observational Astronomy — *October 2024*

Statistics meets ML in Particle Physics and Astrophysics — PHYSTAT
Theme - ML & Astrophysics — *September 2024*

Rubin Community Workshop 2024 — Legacy Survey of Space and Time
Theme - Rubin Science — *July 2024*

Second Intermediate Python for Astronomical Software Development — TVS Science Collaboration LSST
Theme - Git, Software Development — *July 2024*

Utrecht Workshop on Philosophy of Cosmic Inflation — Utrecht University
Theme - Philosophy of Astronomy and Cosmology — *May 2024*

LSST ToO Workshop 2024 — Legacy Survey of Space and Time
Theme - Gravitational Waves, neutrinos, solar system object detection — *March 2024*

CONFERENCES ATTENDED

Transients From Space — Space Telescope Science Institute (STScI)
Theme - Transients — *March 2025*

27th International Microlensing Conference — South African Astronomical Observatory
Theme - Microlensing — *January 2025*

Nancy Grace Roman Space Telescope Summer Conference 2023 — NASA
Theme - Space Telescopes and Cosmic Exploration — *June 2023*

SUPERVISED FORMAL RESEARCH EXPERIENCE

Primordial Non-Gaussianity as a window to New Age Cosmology
Undergraduate Thesis — August 2023 – May 2024

Mechanical Model for Tracking the Optimized Foraging Pattern of Weakly Electric Fish Under Varying Conductivity
Behavioral Ecology — January 2023 – May 2023

Effect of Induction Heating on Temperature and Force Characteristics in Rotary Friction Welding
Metallurgy and Mechanical Applications — December 2022 – December 2023

Implementation of Scheduling Algorithms
Independent Project Course — July 2022 – June 2023

IoT-based Air Quality Sensor Network
Atmospheric Physics — March 2022 – August 2022

CERTIFICATIONS

- R Programming
- Matlab Fundamentals
- Control Design Onramp
- Machine Learning Onramp
- Gravitational Wave Open Data Workshop 6
- Gravitational Wave Open Data Workshop 7

VOLUNTEERING

Khoj Museum | *Environment* January 2021 – April 2021

- As a volunteer, I worked on ideas for sustainability under the domain of water conservation with the sub-domain of Technology and Innovation.
- I made a newsletter, came up with ideas for water sustenance, helped in developing a game based on water sustenance, came up with new technologies for water sustenance, and created activities for the same.